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Ch.10 Human Respiratory System

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. During inhalation, what does the diaphragm do?

Answer: (b) Contracts and moves downward — Diaphragm contraction flattens it, increasing thoracic volume.

2. Which part acts as the respiratory surface?

Answer: (d) Alveoli — Gas exchange occurs across alveolar walls.

3. How many oxygen molecules can bind to one haemoglobin molecule?

Answer: (d) 4 — Each of the four haem groups binds one O_2.

4. Which statement about respiratory pigments is true?

Answer: (a) Transport oxygen from lungs to tissues — Haemoglobin carries O_2 to tissues; CO_2 transport is mainly as bicarbonate.

5. Which respiratory pigment is found in muscle?

Answer: (c) Myoglobin — Myoglobin stores oxygen in muscle tissue.

6. What is the maximum amount of air that can be moved in a maximal respiratory effort?

Answer: (b) Vital capacity — Vital capacity is the maximum air exhaled after maximum inhalation.

7. In what form is most CO_2 transported in blood?

Answer: (c) Converted to bicarbonate ions — About 70% of CO_2 is transported as HCO_3^- in plasma.

8. Which treatment is commonly used for pulmonary TB?

Answer: (a) Antibiotics — TB is treated with a prolonged combination of anti-TB antibiotics.

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9. Which can cause pneumonia?

Answer: (d) All of these — Pneumonia may be caused by bacteria, viruses or fungi.

10. Emphysema is characterized by what?

Answer: (c) Destruction of alveoli — Alveolar walls break down, reducing surface area for gas exchange.

SECTION 2: SHORT QUESTIONS

1. Define respiratory surface and list its properties.

A respiratory surface is the area where gas exchange occurs between the environment and blood. It should be thin, moist, permeable, well supplied with blood, structurally supported so it stays open, located internally to reduce water loss, ventilated by air movement, and reached through branched airways that humidify the air.

2. How does the nasal cavity filter inhaled air?

Hairs trap large particles; mucus traps dust/microbes; cilia move mucus toward the pharynx; rich blood supply warms and moistens air.

3. Trace the path of air through the respiratory system.

> Nasal cavity → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli

4. Describe the structure and function of alveoli.

Alveoli are thin-walled air sacs surrounded by capillaries. Their moist epithelium and large surface area allow rapid diffusion of O_2 into blood and CO_2 out of blood.

5. What is the role of the diaphragm in inhalation and exhalation?

During inhalation it contracts and flattens, increasing thoracic volume. During exhalation it relaxes and domes upward, decreasing volume and helping expel air.

6. What are the three ways CO_2 is transported in blood?

As bicarbonate ions (major), bound to haemoglobin as carbaminohaemoglobin, and dissolved in plasma.

7. Why are millions of alveoli better than two simple balloon-like lungs?

Millions of alveoli provide a huge surface area, short diffusion distance and rich capillary supply, making gas exchange much more efficient.

8. Differentiate the following pairs.

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i. Internal and external respiration

External respiration	Internal respiration
Site: Lungs/alveoli ↔ blood.	**Site:** Blood ↔ body tissues.
Gas movement: O_2 into blood, CO_2 into alveoli.	**Gas movement:** O_2 into tissues, CO_2 into blood.

ii. Upper and lower respiratory tract

Upper tract	Lower tract
Parts: Nasal cavity, pharynx, larynx.	**Parts:** Trachea, bronchi, bronchioles, lungs.
Main role: Filter, warm, moisten air.	**Main role:** Conduct air and exchange gases.

iii. Bronchi and bronchioles

Bronchi	Bronchioles
Structure: Larger tubes with cartilage.	**Structure:** Smaller tubes without cartilage.
Function: Conduct air into lungs.	**Function:** Distribute air to alveoli; regulate airflow.

iv. Haemoglobin and myoglobin

Haemoglobin	Myoglobin
Location: Red blood cells.	**Location:** Muscle cells.
Role: Transports O_2 in blood.	**Role:** Stores O_2 in muscle.
O_2 affinity: Lower than myoglobin.	**O_2 affinity:** Higher; releases O_2 at low tissue pO_2.

LONG QUESTIONS

1. Describe the mechanism of inhalation and exhalation.

Inhalation (inspiration)

Diaphragm contracts and flattens downward → External intercostal muscles contract, lifting ribs → Thoracic volume increases → Lungs expand (pleural membranes adhere) → Intrapulmonary pressure falls below atmospheric → Air flows into lungs

Exhalation (expiration)

Diaphragm relaxes and domes upward → Intercostal muscles relax, ribs move down/in → Thoracic volume decreases → Elastic recoil of lungs → Intrapulmonary pressure rises above atmospheric → Air flows out

Breathing is controlled by the respiratory centre in the medulla oblongata. Forced breathing uses additional accessory muscles.

2. Describe the transport of oxygen through blood.

Oxygen diffuses from alveoli ($pO_2 \approx 105$ mm Hg) into pulmonary capillaries. A small amount dissolves in plasma (about 3 mL O_2/L), but most binds to haemoglobin in RBCs, giving whole blood a capacity of about 200 mL O_2/L.

- About 97% of haemoglobin becomes oxyhaemoglobin in the lungs (bright red).
- In tissues, lower pO_2 (≈40 mm Hg at rest), higher CO_2, lower pH and higher temperature reduce haemoglobin's affinity for O_2 (Bohr effect), so oxygen is unloaded.
- At rest about 22% of bound O_2 is released; during exercise up to 62% may be released, while a reserve remains.

Thus haemoglobin loads O_2 efficiently in lungs and unloads it where metabolism demands it.

3. Describe the transport of carbon dioxide through blood.

Form	Approx. share	Mechanism
Bicarbonate ions (HCO_3^-)	~72%	$CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic anhydrase in RBCs) → $H^+ + HCO_3^-$; HCO_3^- moves to plasma via chloride shift.
Carbaminohaemoglobin	~20%	CO_2 binds to globin chains of haemoglobin.
Dissolved in plasma	~8%	CO_2 dissolves directly in plasma water.

In the lungs, low alveolar pCO_2 reverses the reactions: HCO_3^- re-enters RBCs, combines with H^+ to form H_2CO_3, which splits into CO_2 and H_2O; CO_2 diffuses into alveoli and is exhaled.

The textbook uses the term "carboxyhaemoglobin" for CO_2-bound haemoglobin; the correct term is carbaminohaemoglobin (carboxyhaemoglobin refers to CO-bound haemoglobin).

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4. Describe the structure and function of haemoglobin.

Haemoglobin is a quaternary protein in RBCs composed of four globin polypeptide chains (two α chains of 141 amino acids and two β chains of 146 amino acids) and four haem groups.

- Each haem group contains an Fe^{2+} ion held in a porphyrin ring; each Fe^{2+} binds one O_2 molecule, so one haemoglobin carries up to four O_2.
- The chains are held together by salt bridges, hydrogen bonds and hydrophobic interactions.
- Cooperative binding gives a sigmoid O_2-dissociation curve: binding of one O_2 increases affinity for the next.

Function: transport O_2 from lungs to tissues and assist CO_2 transport and buffering. Myoglobin, a single-chain muscle pigment, stores O_2 and has higher affinity.

5. Describe causes, symptoms and treatment of sinusitis.

Sinusitis is inflammation of the lining of the paranasal sinuses (four paired air-filled spaces around the nasal cavity). It may be acute (7–10 days) or chronic (>12 weeks).

- **Causes:** most cases are viral; some bacterial; rarely fungal.
- **Symptoms:** fever, plugged nose, pus-like nasal discharge, loss of smell, facial pain, post-nasal drip, headache worsened by bending over.
- **Treatment:** viral cases resolve without antibiotics; bacterial cases may need antibiotics/sulpha drugs; nebulization reduces inflammation; chronic/recurring cases may need nasal surgery.

6. Describe causes, symptoms and treatment of pneumonia and pulmonary tuberculosis.

Pneumonia

- **Cause:** acute infection of alveoli, most commonly by viruses or bacteria (e.g., Streptococcus pneumoniae), less commonly fungi; alveoli fill with pus/fluid.
- **Symptoms:** cough with phlegm, shortness of breath, chest pain, fever, blueness of skin, loss of appetite, high heart rate, fatigue.
- **Treatment:** specific antibiotics for bacterial pneumonia; analgesics for fever/pain; vaccination prevents certain bacterial and viral pneumonias.

Pulmonary tuberculosis

- **Cause:** chronic infection by Mycobacterium tuberculosis; spreads through cough/sneeze; damages alveoli and may form cavities.
- **Symptoms:** cough with blood, intermittent evening fever, night sweats, weight loss, anorexia, weakness, chest pain.
- **Treatment:** multiple antibiotics taken regularly over a long period (about 9 months).

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7. Describe causes, symptoms and treatment of emphysema.

Emphysema is a type of COPD in which the inner walls of alveoli are damaged and rupture, creating fewer, larger air spaces and reducing the surface area for gas exchange.

- **Causes:** primarily smoking; long-term exposure to air pollution, dust or chemical fumes; genetic deficiency of alpha-1 antitrypsin.
- **Symptoms:** shortness of breath, coughing, wheezing, fatigue and chest tightness.
- **Treatment:** quitting smoking is most important; bronchodilators, inhaled steroids, oxygen therapy and pulmonary rehabilitation help manage symptoms; severe cases may need surgery.

Loss of elastic recoil and surface area makes expiration difficult and traps air in the lungs.

INQUISITIVE QUESTIONS

1. How does alveolar structure optimize gas exchange?

Alveoli provide a large surface area, one-cell-thick walls, moist lining and dense capillaries. Short diffusion distance and maintained concentration gradients make O_2/CO_2 exchange rapid.

2. How does COPD affect gaseous exchange efficiency?

COPD narrows airways and/or destroys alveoli, reducing airflow and surface area. Air trapping and poor ventilation lower O_2 uptake and CO_2 removal.

3. Explain external versus internal respiration in gas exchange.

External respiration exchanges gases between alveoli and blood; internal respiration exchanges gases between blood and tissues. Both occur by diffusion along partial-pressure gradients.

4. How does oxygen transport support cellular respiration?

Haemoglobin delivers O_2 to mitochondria, where it acts as the final electron acceptor in aerobic respiration, allowing efficient ATP production.

5. What environmental factors influence gaseous exchange in humans?

Altitude (low pO_2), air pollution/smoking, temperature, humidity, dust/allergens and occupational irritants can affect ventilation, diffusion and airway resistance.

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